

Fluoride Adsorption Hybrid Model

Status: Phase 3 Complete | Phase 4 Ready | 37.5% Progress

Current Performance: Langmuir $R^2 = 0.8158$ | Target Hybrid $R^2 = 0.94$

A hybrid physics-machine learning model for fluoride removal prediction combining Langmuir adsorption theory with data-driven residual correction.

Quick Start

View Project Status

```
cat PROJECT_STATUS.md      # Complete overview with all metrics
cat PHASE_4_EXECUTION.md   # Ready-to-execute Phase 4 guide
```

Run Current Phases (Already Complete)

```
# Phase 1: Design of Experiments
python3 src/phase1_doe_design.py
python3 src/phase1_simulate_adsorption.py

# Phase 2: Langmuir Baseline Model
python3 src/phase2_langmuir_fitting.py

# Phase 3: Residual Analysis (outputs ready in results/phase3/)
# No re-run needed - outputs already verified

# Phase 4: ML Training (NEXT STEP)
# Script template in PHASE_4_EXECUTION.md
```

Verify Data Quality

```
python3 << 'EOF'
import pandas as pd
train = pd.read_csv('results/phase3/ml_training_data.csv')
print(f"Training data: {train.shape} | Test data shape: (100, 39)")
print(f"Residual mean: {train['residual'].mean():.6f} | Std: {train['residual'].std():.4f}")
EOF
```

Project Architecture

```
fluoride-adsorption-hybrid/
  src/
    phase1_doe_design.py      Latin Hypercube Sampling (500 samples)
    phase1_simulate_adsorption.py Physics-based simulation
```

```

phase2_langmuir_fitting.py      Polynomial Langmuir model (R²=0.8158)
[phase4_ml_training.py - TO CREATE]

results/
  phase2/                        Langmuir outputs
    langmuir_predictions.csv
    langmuir_diagnostics.png
    langmuir_model_info.json
  phase3/                        ML training data (ready for Phase 4)
    ml_training_data.csv        (400 × 39)
    ml_test_data.csv            (100 × 39)
    feature_importance.csv
    phase3_diagnostics.png
  phase4/                        ML models (empty - ready for Phase 4)

data/
  raw/                          Reference data
  processed/                     500-sample dataset

Documentation
  PROJECT_STATUS.md              START HERE - Full project overview
  PHASE_4_EXECUTION.md           Ready-to-run Phase 4 guide
  requirements.txt               Dependencies
  README.md (this file)

```

Key Metrics

Phase	Component	Metric	Status
1	DoE Design	500 LHS samples, 10 factors	Complete
2	Langmuir Model	$R^2 = 0.8158$, RMSE = 0.5278 mg/g	Complete
3	Residual Analysis	28 features engineered, pH = 21.1% importance	Complete
4	ML Models	Target: $R^2 \geq 0.70$ on residuals	Ready to Start
5	Hybrid Integration	Expected: $R^2 \geq 0.94$	Planned

Phase Progress

Phase 1: Research & Experimental Design

- 10 factors selected (pH, C , time, dose, temp, flow, chloride, hardness, carbonate, NOM)

- 500-point Latin Hypercube Sampling generated
- Physics-based simulation of fluoride adsorption (corrected for realistic data)

Phase 2: Langmuir Baseline Model

- $R^2 = 0.8158$ — Excellent fit to physics-based data
- 66 polynomial features (10 original + 10 squared + 45 interactions)
- Residuals analyzed: mean = 0 (unbiased), systematic pH-driven pattern identified

Phase 3: Residual Analysis & Feature Engineering

- Residual distribution analyzed by pH: +0.649 mg/g underestimation at pH 6.5–7 (optimal)
- 28 engineered features created (6 groups: pH deviation, ion competition, equilibrium, etc.)
- Quick RandomForest baseline: 47.3% test R^2 on residuals (proof of learnability)
- ML training/test data prepared: 400/100 split, 38 features, ready for Phase 4

Phase 4: ML Model Training (NEXT)

- Train RandomForest, XGBoost, MLP on Phase 3 engineered features
 - Target: 70% residual $R^2 \rightarrow$ enables hybrid model 94% overall
 - Input: 400 training + 100 test samples (verified, clean)
 - Expected duration: 2–3 hours
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Next Steps

Immediate: Execute Phase 4

1. Review `PHASE_4_EXECUTION.md` for complete setup
2. Create `src/phase4_ml_training.py` using provided template
3. Run: `python3 src/phase4_ml_training.py`
4. Verify outputs in `results/phase4/`

Timeline

- Phase 4 execution: ~2–3 hours
 - Phase 5 (Hybrid integration): ~1 hour
 - Total to completion: ~3–4 hours
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File Registry

Phase 2 (Langmuir) Outputs

- `results/phase2/langmuir_predictions.csv` — 500 samples with Langmuir predictions
- `results/phase2/langmuir_model_info.json` — Model parameters and metrics
- `results/phase2/langmuir_diagnostics.png` — 4-panel diagnostic plots

Phase 3 (Residual Analysis) Outputs

- `results/phase3/ml_training_data.csv` — 400 samples with 38 features (ready for ML)
 - `results/phase3/ml_test_data.csv` — 100 samples with 38 features (evaluation set)
 - `results/phase3/feature_importance.csv` — Feature importance rankings
 - `results/phase3/residual_analysis.json` — Metadata (skewness, means, etc.)
 - `results/phase3/phase3_diagnostics.png` — Residual analysis plots
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Scientific Foundation

10 Factors Selected (Phase 1)

Factor	Range	Unit	Justification
pH	3–9	—	Controls surface charge & F speciation
C (Initial F)	10–100	mg/L	Driving force for adsorption
Contact Time	5–180	min	Kinetic equilibration
Adsorbent Dose	1–50	g/L	Binding capacity
Temperature	20–60	°C	Thermal effects
Flow Rate	0.5–2.5	mL/min	Transport mechanism
Chloride	0–100	mg/L	Competitive ion
Hardness (Ca ²)	0–200	mg/L	Ionic strength
Carbonate	0–50	mg/L	pH buffer effect
NOM	0–50	mg/L	Pore fouling

Physics: Langmuir Isotherm

`q(C, dose, ...) = polynomial(C, time, temp, pH, dose, ...)`

- **R² = 0.8158**: Chemistry explains 81.58% of variance
- **Residuals**: 18.42% variance is learnable, pH-driven pattern

ML Enhancement: Residual Correction

`q_final = q_Langmuir + ML_correction`

Expected: R² 0.94

Environment Setup

Requirements

- Python 3.10+
- scikit-learn 0.24+
- pandas, numpy, scipy, matplotlib
- xgboost (for Phase 4)

Install

```
pip install -r requirements.txt
python3 -m venv venv # If needed
```

Verify Installation

```
python3 -c "import pandas; import sklearn; import xgboost; print(' All dependencies ready')"
```

Expected Final Results

Hybrid Model Performance (Phase 5)

Langmuir (Phase 2): $R^2 = 0.8158$

Residual ML (Phase 4): $R^2 = 0.70$

Hybrid Model: $R^2 = 0.94$ TARGET ACHIEVED

Practical Impact

- **Before:** ± 0.53 mg/g RMSE (7.9% error)
 - **After:** ± 0.25 mg/g RMSE (3.7% error) — **66% improvement**
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Documentation

- **PROJECT_STATUS.md** — Complete metrics, findings, file registry
 - **PHASE_4_EXECUTION.md** — Step-by-step Phase 4 guide with script template
 - **PHASES_1_3_SUMMARY_REPORT.md** — Comprehensive technical report
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Success Criteria

- ☒ Phase 1: 500-sample LHS design generated
 - ☒ Phase 2: Langmuir model $R^2 = 0.80$
 - ☒ Phase 3: Residual patterns identified and features engineered
 - ☐ Phase 4: ML models trained with $R^2 = 0.70$ on residuals
 - ☐ Phase 5: Hybrid model verified with $R^2 = 0.94$
 - ☐ Phase 6: External validation completed
 - ☐ Phase 7: Documentation finalized
 - ☐ Phase 8: Model deployed
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Quick Links

- **Start here:** PROJECT_STATUS.md

- **Next execution:** PHASE_4_EXECUTION.md
- **Technical details:** PHASES_1_3_SUMMARY_REPORT.md
- **Results location:** results/phase{2,3}/

Last Updated: May 6, 2026

Python Version: 3.13

Project Status: Phase 3 Complete | Awaiting Phase 4 Execution